

Worked Solution (Formula Solution)

a. What is the present value of \$12,000 received 14 years from today when the interest rate is 4% per year?

The present value of a cash flow can be found using the following formula:

$$PV = \frac{C}{(1 + r)^n}$$

where PV is the present value, C is the cash flow, r is the interest rate, and n is the number of periods.

The present value is computed using the following formula:

$$PV = \frac{\$12,000}{(1 + 0.04)^{14}} = \$6,929.70$$

The present value of \$12,000 received 14 years from today when the interest rate is 4% per year is \$6,929.70.

b. What is the present value of \$12,000 received 20 years from today when the interest rate is 9% per year?

The present value is computed using the following formula:

$$PV = \frac{\$12,000}{(1 + 0.09)^{20}} = \$2,141.17$$

The present value of \$12,000 received 20 years from today when the interest rate is 9% per year is \$2,141.17.

c. What is the present value of \$12,000 received 7 years from today when the interest rate is 2% per year?

The present value is computed using the following formula:

$$PV = \frac{\$12,000}{(1 + 0.02)^7} = \$10,446.72$$

The present value of \$12,000 received 7 years from today when the interest rate is 2% per year is \$10,446.72.

